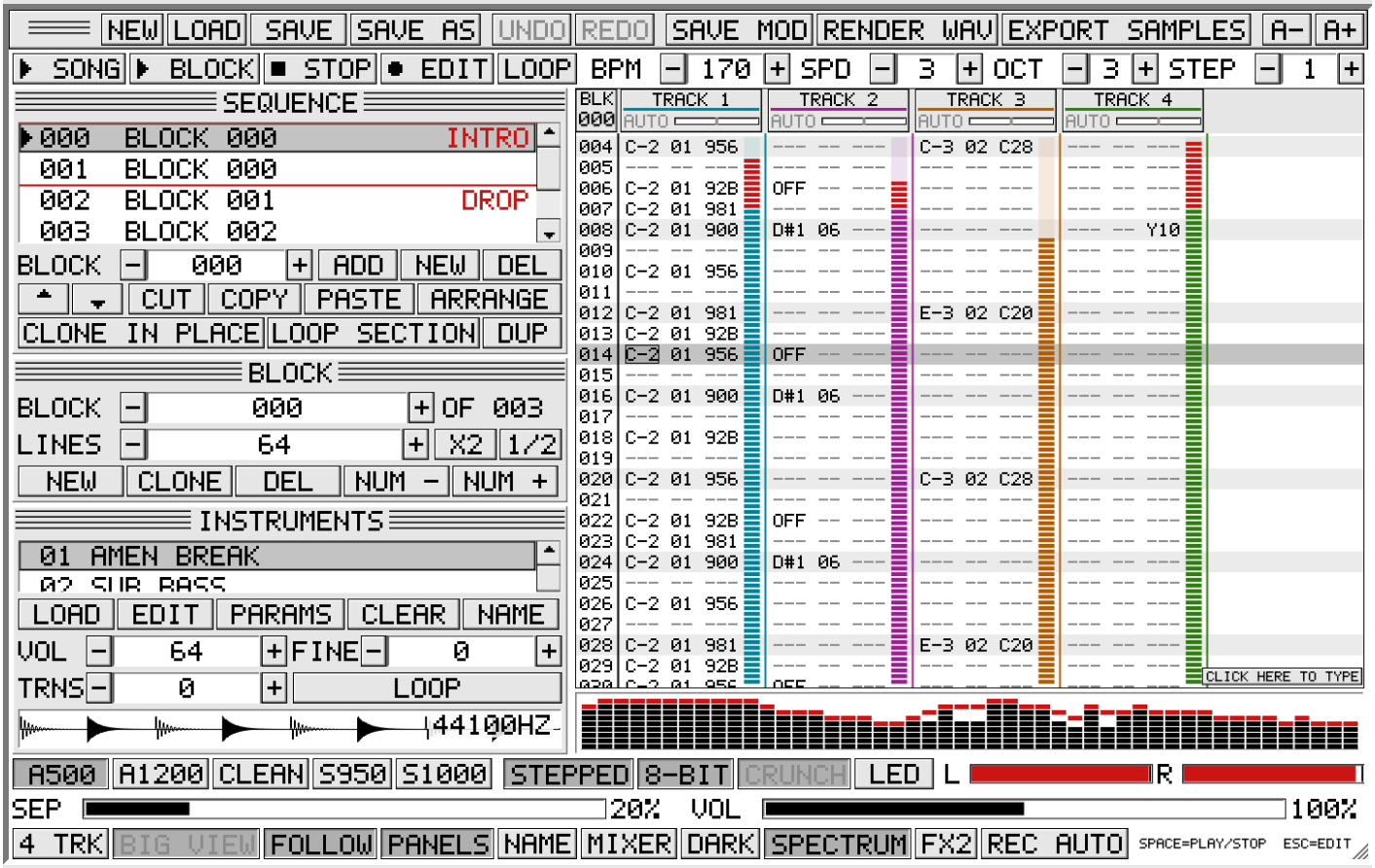
PAULA'S REVENGE V0.12

USER MANUAL

AN 8-TRACK AMIGA-STYLE SAMPLE TRACKER FOR JUNGLE, DRUM & BASS AND BREAKS



THE MAIN SCREEN, LIGHT SKIN

Paula's Revenge is an original program inspired by the classic Amiga trackers of the early 90s. Paula and Akai sound, a mixer, plug-ins, MIDI out, automation, bouncing and stems.

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GETTING STARTED

Paula's Revenge is an 8-track sample tracker for jungle, drum & bass and anything else that wants the crunchy sound of the Amiga's Paula chip and the old Akai samplers. You write music as lines of notes and commands that play from top to bottom, one block at a time, in the order the song lists them.

INSTALL

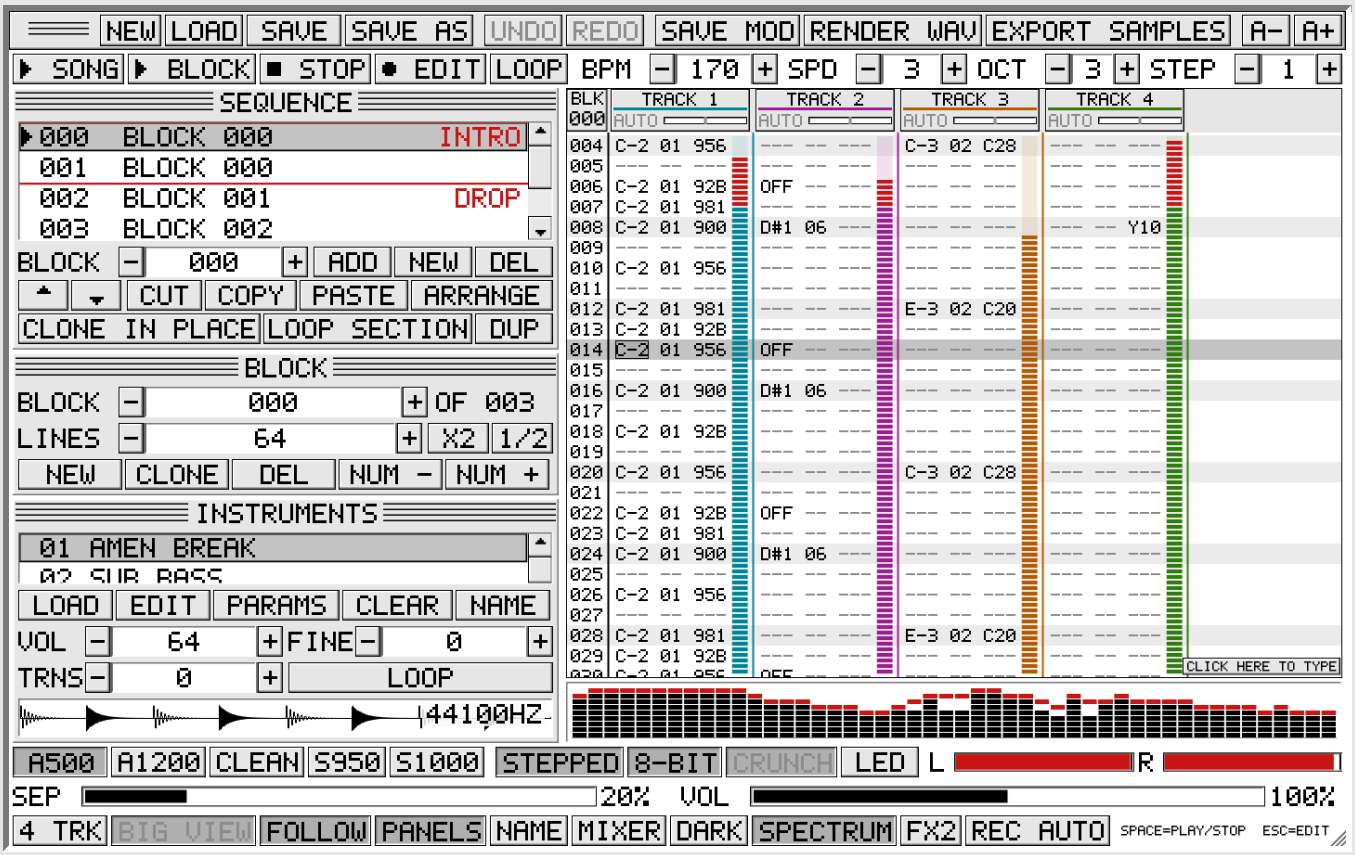
1. Open **Releases** on the right of the GitHub repo page and download `PaulasRevenge-macOS.zip` from the newest one.
2. Follow `INSTALL.txt` inside: copy the app into Applications and clear the macOS quarantine.
3. Open it. Audio and MIDI devices are in the menu under **Audio/MIDI Settings…** (CMD ,).

YOUR FIRST SOUND IN A MINUTE

1. Press SPACE. A new song already has a few chip sounds in it, so the keys make sound straight away.
2. Press ESC to switch • EDIT on. The cursor line turns grey.
3. Type notes with the bottom two rows of the keyboard (Z is C, S is C#, X is D…). Each note goes in with the current instrument and the cursor moves down by STEP.
4. Click LOAD in the instrument panel to pick a break from your drive. Every sample you highlight plays straight away.
5. Press SPACE to hear the block. SHIFT SPACE plays the whole song.

TIP For a break: put `C-3` on line 0 and use `9xx` offsets to jump around inside it, or slice it in the sample editor and play the slices from the keys.

THE SCREEN



THE MAIN SCREEN, LIGHT SKIN, PLAYING

AREA	WHAT IT DOES
Title bar	NEW, LOAD, SAVE, SAVE AS, UNDO, REDO, SAVE MOD, RENDER WAV, EXPORT SAMPLES, and text size A- A+.
Transport	Play the song or the block, stop, EDIT mode, LOOP (the marked lines), BPM, SPD (ticks per line), octave and step.
Sequence	The song order. Click a position to go there, drag to highlight positions, and set which block plays where.
Block	Pick, add, clone or delete blocks; set the number of lines (1–256); X2 doubles a block, 1/2 halves it.
Instruments	63 slots. LOAD, EDIT, PARAMS, the instrument's VOL, finetune, transpose, loop, and a little waveform.
Pattern	4 or 8 tracks. Click a track name to mute it, right-click to solo; drag in the bar under the name to pan the track.
Output strip	The machine you hear (A500, A1200, CLEAN, S950, S1000), its options, an L/R meter, stereo separation and volume.

AREA	WHAT IT DOES
Bottom row	4 or 8 tracks, BIG VIEW, FOLLOW, PANELS, song NAME, MIXER, DARK, SPECTRUM, FX2, REC AUTO, and the status line.

The app opens with the dark skin. DARK switches to black on white, as in this manual. BIG VIEW shows 4 tracks at a time with large text: TAB to track 5 and the view flips to tracks 5–8.

The status line at the bottom right says what just happened, and what the command under the cursor does.

WRITING PATTERNS

Each track has a note, an instrument number and an effect command per line: `C-2 01 956` means note C-2, instrument 01, command `9` with value `56`. A second effect column appears when you switch `FX2` on.

NOTES AND THE CURSOR

- With `• EDIT` on, the bottom rows of keys play and enter notes from the octave in `OCT`; the `Q` row is the octave above. `1` enters a note-off (`OFF`).
- In the instrument and command columns, type hex digits `0`–`9` `A`–`F`, or a letter `G`–`Z` for the extra commands.
- `DELETE` clears the field under the cursor. `SHIFT` `DELETE` removes the line and pulls the track up; `ALT` `DELETE` inserts a blank line.
- A MIDI keyboard plays the current instrument and enters notes when EDIT is on. MIDI 60 = `C-3`.

HIGHLIGHTING WITH THE MOUSE

Drag across the pattern to highlight notes and effects; it snaps to whole fields and can cover several tracks. `SHIFT`-click highlights from the cursor to the click, and `CMD` `A` highlights the whole block.

- `CMD` `C` / `CMD` `X` copy or cut the highlight; `CMD` `V` pastes at the cursor, each part into the same kind of field. Without a highlight they work on the whole track (add `SHIFT` for the block).
- `CMD` `↑` / `CMD` `↓` transpose the track a semitone, with `SHIFT` an octave.
- Right-click inside a highlight to bounce it to a new sample.

LOOPING SOME LINES

Drag down the line numbers to mark lines; they turn red and playback goes round them. `CMD` `[` and `CMD` `]` set the start and end at the cursor. `LOOP` on the transport row turns the loop off and on again.

PLAYING A BLOCK

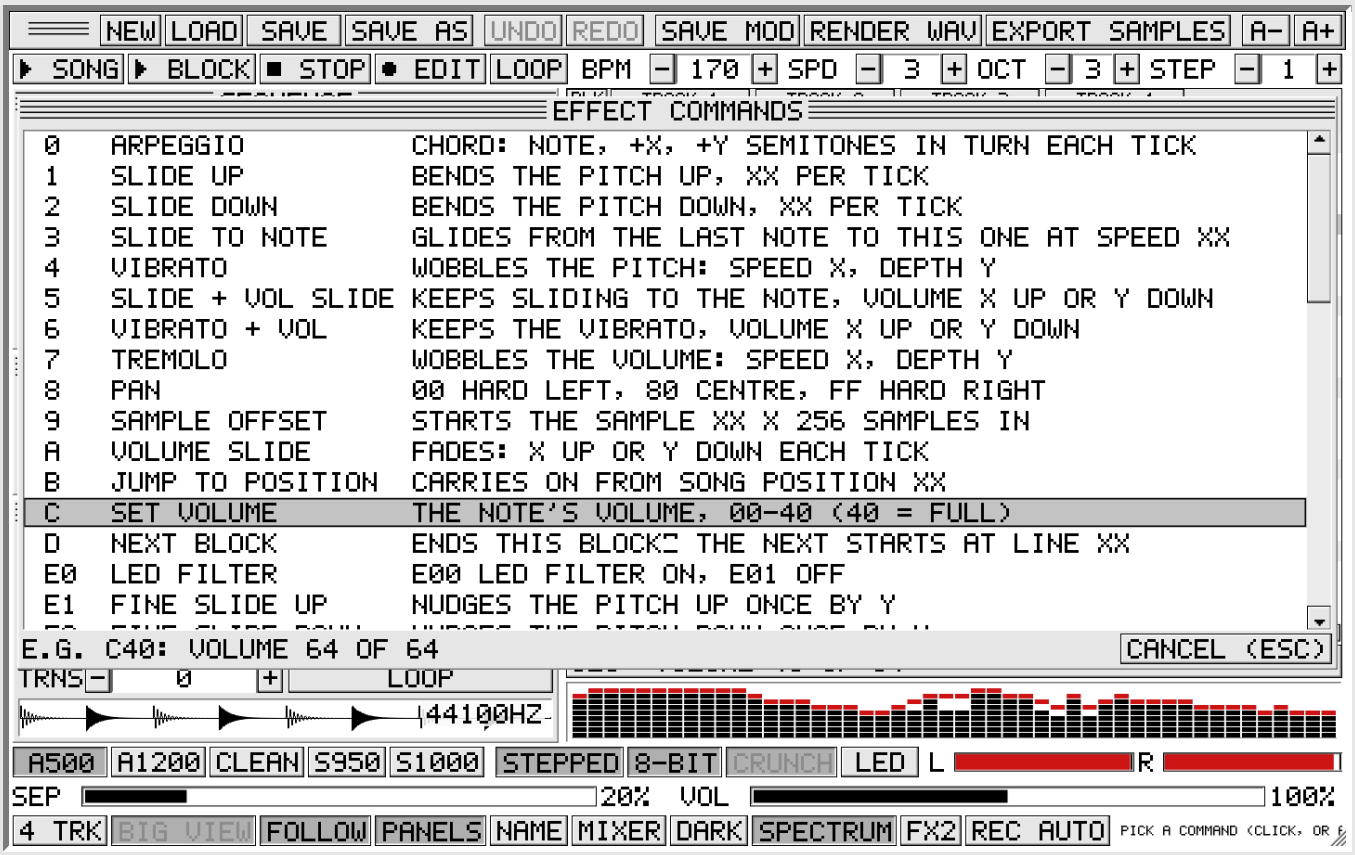
While a block plays, picking another block (in the sequence, the BLOCK box or the BLK corner) carries playback over to it on the same line, without stopping, so you can try blocks against each other in time.

UNDO

One history covers everything: notes, blocks, the sequence, instruments, samples, the mixer and song settings. CMD Z and CMD SHIFT Z work on every page, and the status line says what was undone. It keeps the last 200 steps.

EFFECT COMMANDS

Right-click an effect cell for a list of every command with what it does. Click one, or type its letter, and it goes into the cell with a useful starting value, the cursor ready on the value. Put the cursor on any command to see it explained in the help strip under the pattern.



RIGHT-CLICK AN EFFECT CELL: THE COMMAND LIST

CLASSIC COMMANDS (0–F)

The same numbering as ProTracker, so MOD files load straight in.

CMD	EFFECT
0xy	Arpeggio (+x, +y semitones). On a plug-in or MIDI-out instrument with CHORD on, a held chord instead.
1xx 2xx	Slide pitch up / down
3xx	Slide to the note
4xy	Vibrato (speed x, depth y)

CMD	EFFECT
`5xy` `6xy`	Slide to note / vibrato, plus volume slide
`7xy`	Tremolo
`8xx`	Pan: `00` left, `80` centre, `FF` right
`9xx`	Sample offset (xx X 256)
`Axy`	Volume slide (x up, y down)
`Bxx`	Jump to song position xx
`Cxx`	Volume (`00` - `40`); velocity on plug-in and MIDI-out instruments
`Dxx`	Next block, starting at line xx
`E0x`	LED filter: `E00` on, `E01` off
`E1x` `E2x`	Fine slide up / down
`E4x` `E7x`	Vibrato / tremolo waveform
`E5x`	Finetune
`E6x`	Loop: `E60` marks the start, `E6x` repeats x times
`E8x`	Coarse pan
`E9x`	Retrigger every x ticks
`EAx` `EBx`	Fine volume up / down
`ECx`	Cut the note at tick x
`EDx`	Delay the note by x ticks
`EEx`	Delay the line by x lines
`Fxx`	`01` - `1F` speed, `20` - `FF` BPM, `00` stop

EXTRA COMMANDS (G-Z)

Type the letter in an effect column, then two hex digits.

CMD	EFFECT
`Gxx`	Glide to the note over xx ticks

CMD	EFFECT
Hxx Lxx Wxx	High-pass / low-pass / band-pass filter, cutoff xx (00 20 Hz – FF 20 kHz). On a plug-in or MIDI-out instrument, its learned knobs instead.
Qxx	Filter resonance (or a learned knob)
Ixx	Transpose just this note: 80 none, 8C +12, 74 -12
Kxx	Chance: the step plays xx/FF of the time
Mxx	Micro-move: 01–7F late by xx/128 of a line, 81–FF early
Nxx	Note length: release after xx ticks
Oxx	Overdrive (on plug-in / MIDI-out instruments: the mod wheel)
Pxx	Start at xx/256 of the sample
Rxy	Roll: retrigger every x ticks, y 1–7 quieter or 9–F louder each time
Sxx	Play slice xx+1
Txx	Micro-tune: 80 = 0, each step 1 cent
U01 U00	Play backwards / forwards
Xxy	Randomise: x 0 note ±y, 1 volume, 2 pan, 3 start
Zxx	Bit crush to xx bits
Vxx Jxx	The track's automation knobs AUTO A and AUTO B (see Automation)
Yxx	The next knob change on this track glides over xx ticks

BLOCKS & THE SONG

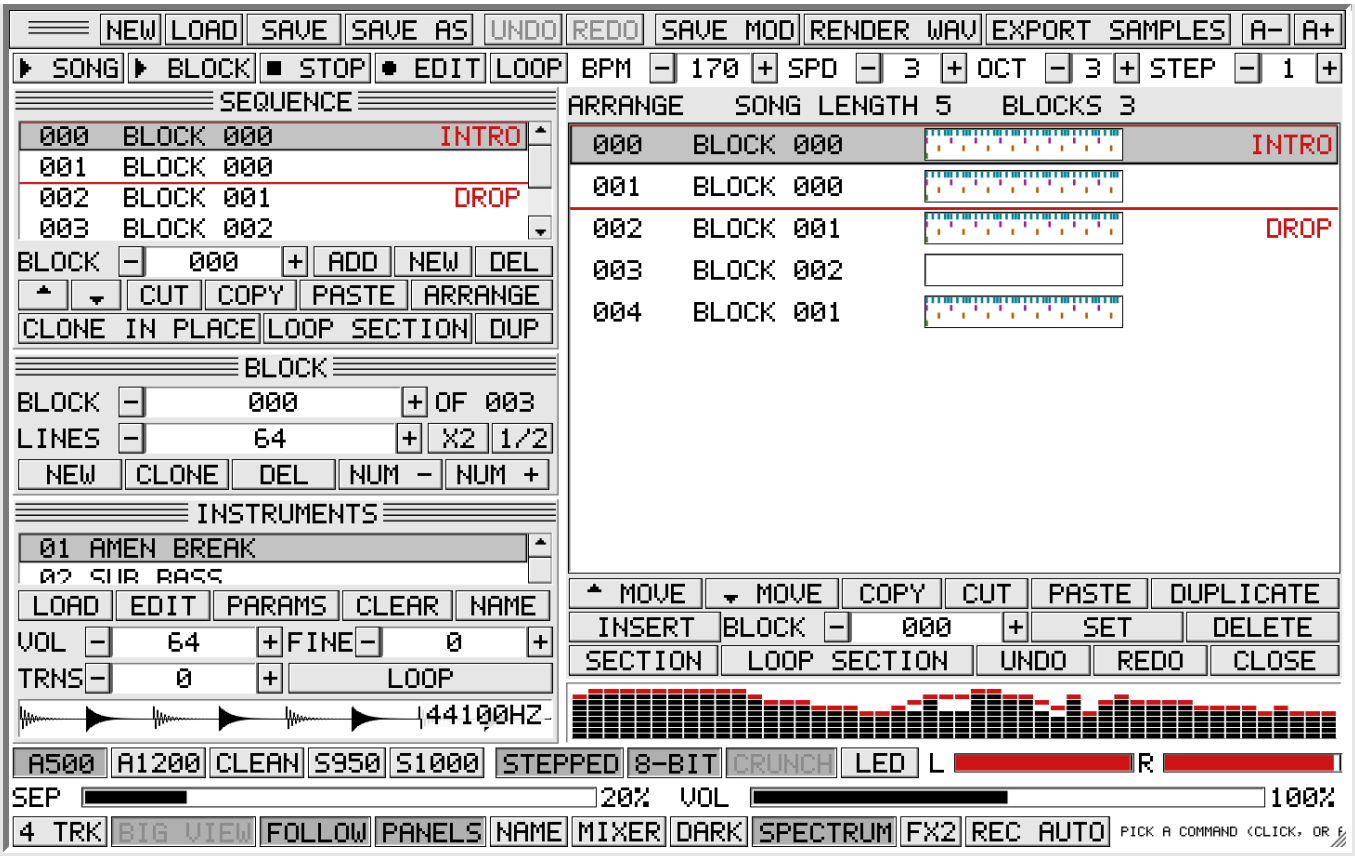
A song is a list of positions, each playing a block. The same block can play at many positions; change it once and every position hears the change.

THE SEQUENCE PANEL

- ADD puts a new position after the selected one, playing the same block; change it with BLOCK - +. NEW makes a new empty block in a new position.
- Drag over positions to highlight them, then CUT COPY PASTE DUP DEL. Drag the highlight, or use ▲ ▼, to move positions.
- CLONE IN PLACE gives a position its own copy of its block, in the next free block, so you can change it without touching the other places it plays.
- Click at the right of a row to start a named **section** there (INTRO, DROP…). LOOP SECTION loops the section you're in; pick a position in another section and playback goes there when the current block ends.

THE ARRANGE PAGE

ARRANGE shows the song order full size, with a picture of each block's notes. It shares the highlight and clipboard with the sequence panel. INSERT adds the BLOCK number after the cursor, SET makes the highlighted positions play it.



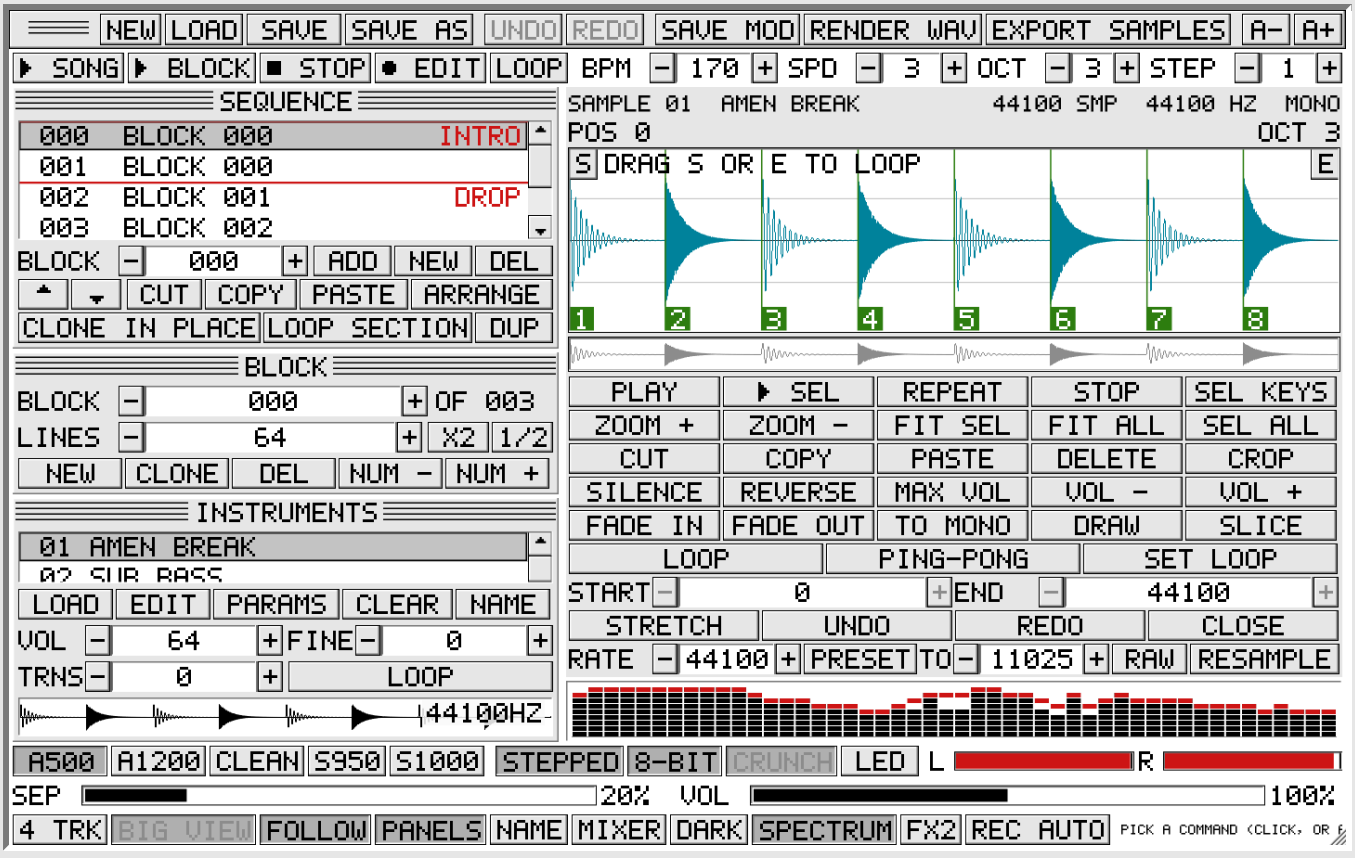
THE ARRANGE PAGE, WITH TWO SECTIONS

THE SAMPLE BROWSER

LOAD opens the browser in place of the pattern. Every sample you highlight plays straight away, and the note keys or a MIDI keyboard play it at other pitches. Shortcut buttons go to Home, Music, Desktop, Downloads, Documents and Drives, and four FAV buttons hold your own folders (click an empty one to set it; SHIFT-click changes it, ALT-click clears it). ENTER or a double-click loads the sample.

THE SAMPLE EDITOR

EDIT (or a double-click on the little waveform) opens the editor.



THE SAMPLE EDITOR, WITH SLICES

- Drag to select; grab either end of a selection to move just that end. The note keys and MIDI play the sample through the full instrument sound; SEL KEYS plays just the selection.
- CUT COPY PASTE DELETE CROP SILENCE REVERSE MAX VOL VOL - VOL + FADE IN FADE OUT TO MONO, and DRAW to draw the wave.

- **Loop:** drag the `S` and `E` handles at the top of the wave; you hear the loop while you drag. `PING-PONG` plays it forwards then backwards.

RATES

Each instrument has a RATE: the speed it plays at on `C-3`. Samples load at their own rate, so they sound as recorded. `RATE` changes it without touching the data (lower rate, lower and slower); `RESAMPLE` converts the data to a new rate and keeps the pitch, with `RAW` for the rough, unfiltered old-sampler way.

SLICING

`SLICE` opens the slicer. `AUTO` finds the hits (`SENS` sets how quiet), `EVEN` cuts equal slices. Drag a tab to move a slice point, double-click the wave to add one, `ALT`-click a tab to remove it. With `SLICE TO KEYS` on, `C-3` plays slice 1, `C#3` slice 2 and so on; without it, `Sxx` picks the slice.

TIME STRETCH

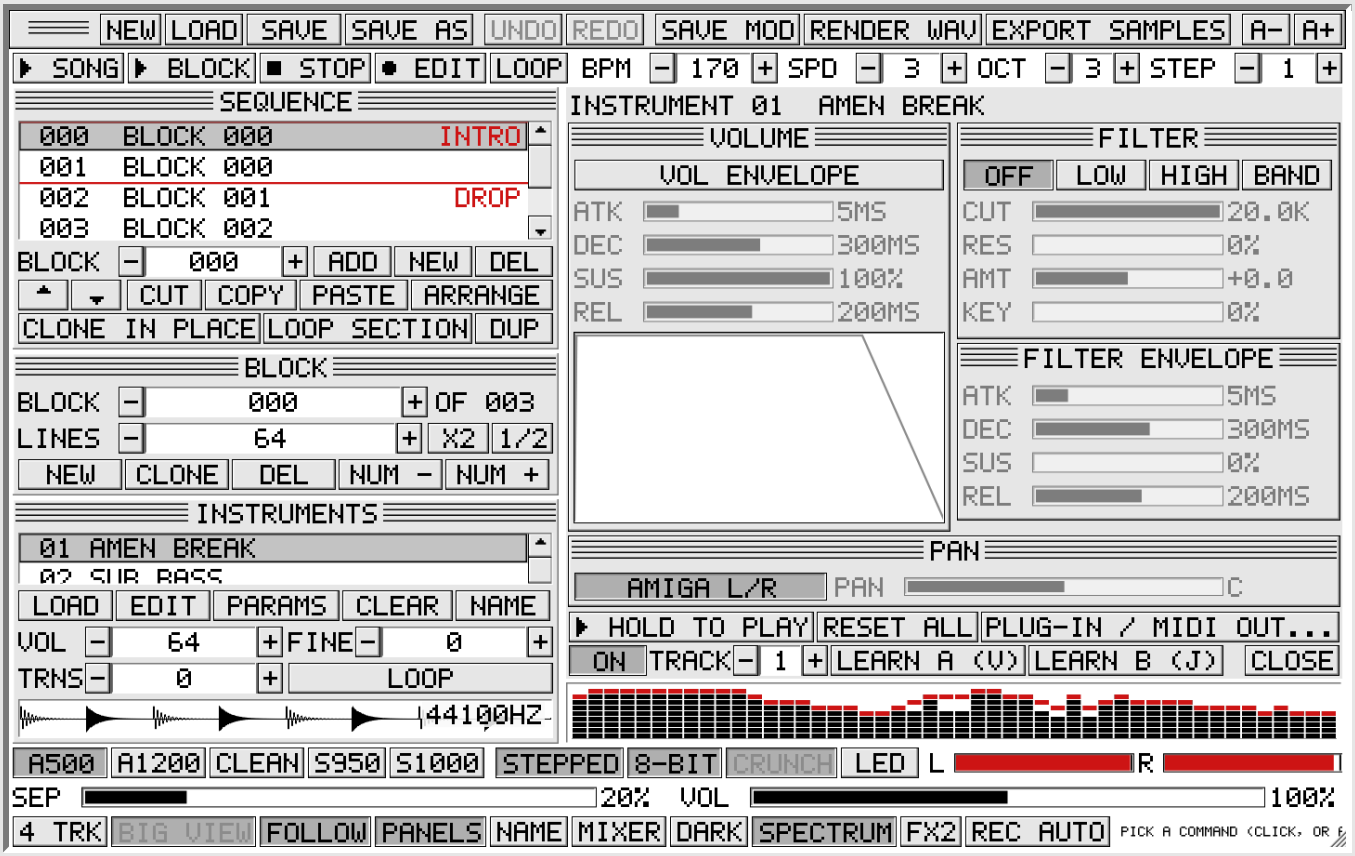
`STRETCH` opens the old-sampler stretch: **CLASSIC** (the Akai sound, matched to recordings of the real thing), **REVISED** (cleaner joins), **HARD** (no blending) and **AKAIZER** (Akaizer v1.2's granular method, rebuilt). Set TIME %, SEMI and CENTS and the CYCLE length; `FIT TO LINES` makes the selection last a number of lines at the song's tempo. `PREVIEW`, then `APPLY`.

INSTRUMENT VOL AND SWITCHING OFF

VOL is a level for the whole instrument on top of each note's `Cxx`. Any instrument can be switched off with its `ON` button (on its PARAMS page): its notes are silent and the list shows it dimmed with `OFF`.

SOUND SHAPING

PARAMS opens the current instrument's page. Everything starts off or neutral, so MODs and older songs sound exactly as they did.



PARAMS: ENVELOPE, FILTER, PAN, AND THE ON BUTTON

- **Volume envelope:** VOL ENVELOPE on, then ATK, DEC, SUS, REL (1 ms – 10 s). With it on, a note-off starts the release instead of cutting the note.
- **Filter:** OFF / LOW / HIGH / BAND, CUT, RES, AMT (how far the filter envelope moves it) and KEY (how far the cutoff follows the note). The filter envelope has its own ATK, DEC, SUS, REL.
- **Pan:** AMIGA L/R follows the Amiga track layout; switch it off to give the instrument its own PAN.
- ▶ HOLD TO PLAY or the keys to hear it; RESET ALL puts everything back; double-click any slider for its default.
- ON switches the instrument off and on. To switch it from the pattern, pick a track, press LEARN A (V) or LEARN B (J), then click ON: that track's V00 switches it off and V01–VFF on.

- PLUG-IN / MIDI OUT... plays the instrument with a VST3 / AU instrument or with your hardware instead (see Plug-ins and MIDI out).

PAULA & THE AKAIS

The output strip picks the machine you hear. None of these change your samples; switch between them any time.

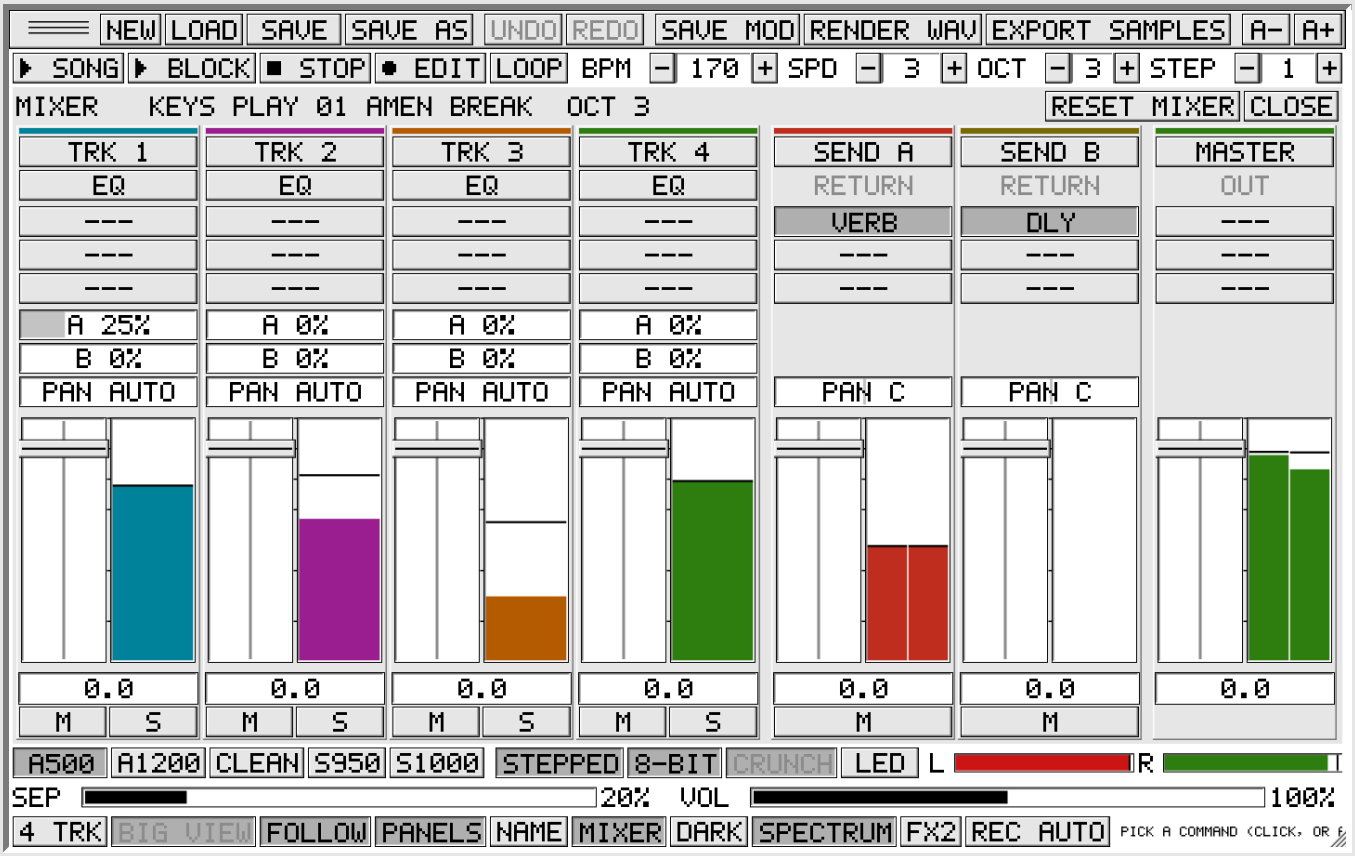
CONTROL	WHAT IT DOES
A500	The original A500's 4.4 kHz output filter: warm and dull
A1200	An open output filter: brighter, more aliasing sparkle
CLEAN	No output filtering
S950	Akai S950: 12-bit, stepped variable-rate playback, its 6-pole filter
S1000	Akai S1000: 16-bit, interpolated playback, clean 44.1 kHz-style output
STEPPED	Sample-and-hold playback like the real chip (off = smooth)
8-BIT	Samples play at 8-bit resolution
CRUNCH	In 8-track songs, track pairs are mixed and squashed back to 8 bits, as 8-channel Amiga trackers did
LED	The Amiga's "LED" low-pass filter at the start of the song (`E00` / `E01` switch it while playing)
SEP	How far tracks lean to the Amiga sides (1 and 4 left, 2 and 3 right). 20% to start; 0% = all centred

PANNING, IN ORDER

1. `8xx` in the pattern (an absolute position, not scaled by SEP).
2. The track's pan, the bar under its name (AUTO = none).
3. The instrument's PAN, when its AMIGA L/R is off.
4. Otherwise Amiga style, by SEP. Stereo samples play centred with their own stereo image.

THE MIXER

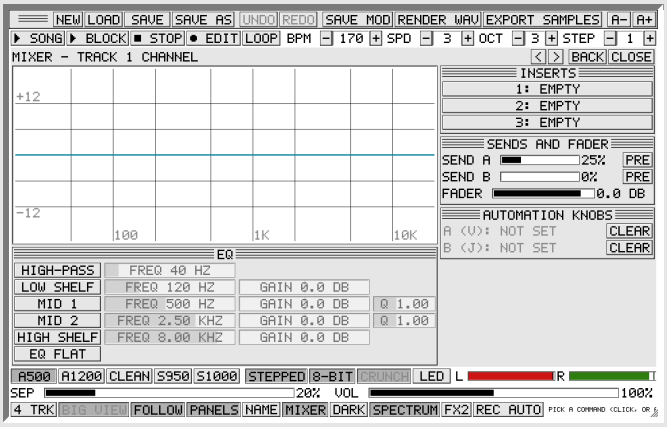
MIXER on the bottom row (CMD 2) opens the mixer across the window; ESC or CMD 1 goes back. The keys still play the current instrument while it's open.



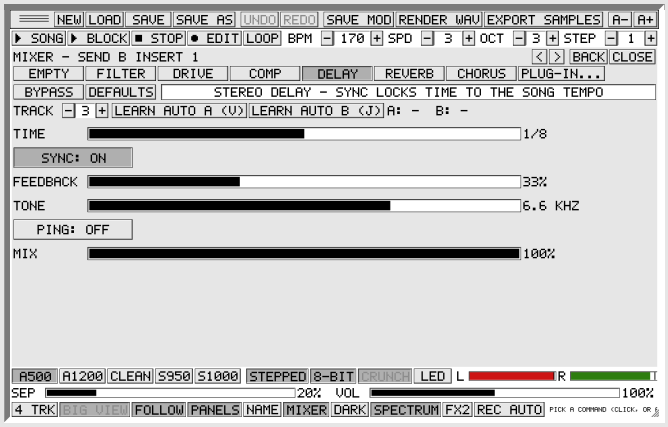
THE STRIPS: TRACKS, SEND A, SEND B, MASTER

STRIP PART	WHAT IT DOES
Name / EQ	Opens the track's channel page. EQ is lit when a band is on; EQ AB when automation knobs are set.
3 insert slots	Click for the effect page; SHIFT-click bypasses (a bypassed slot is dimmed).
A, B	How much goes to SEND A / B. AP / BP = pre-fader. Double-click = off.
PAN	The track's pan. Double-click = AUTO.
Fader	-INF to +6 dB with the meter beside it. SHIFT = fine, double-click = 0 dB.
M / S	Mute and solo.

EQ → INSERTS → PRE SENDS → FADER → POST SENDS → MASTER → PAULA / AKAI OUTPUT



CHANNEL PAGE: EQ, INSERTS, SENDS



EFFECT PAGE: SEND B'S DELAY

- **Channel page:** five EQ bands (HIGH-PASS, LOW SHELF, MID 1, MID 2, HIGH SHELF) with a response curve; click a band's name to switch it on. `<` `>` step through the tracks.
- **Effect page:** FILTER, DRIVE, COMP, DELAY (in ms or synced to the song), REVERB, CHORUS, or `PLUG-IN...`. Every page has `BYPASS` and `DEFAULTS`.
- **Returns:** SEND A starts with a reverb and SEND B with a delay, fully wet, silent until a track sends to them.
- **Delay compensation:** plug-ins that give their sound back late (lookahead limiters, linear-phase EQs) are lined up, so every track stays in time, sends included. The strips page title shows the total delay.

A mixer with nothing changed is skipped, so the song sounds exactly as it would without one. Everything in it is saved in the song, undoable and in renders.

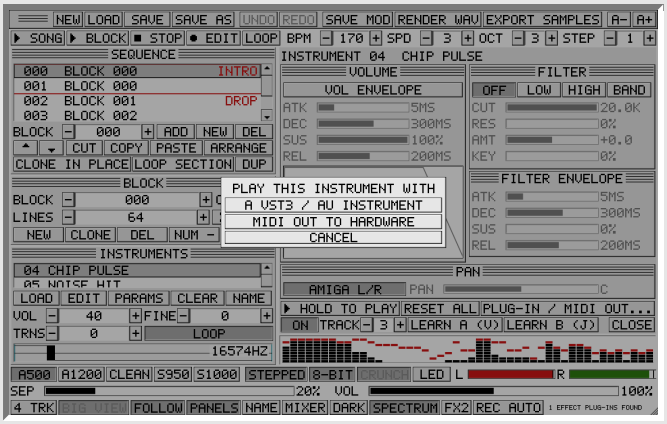
PLUG-INS

EFFECTS IN THE MIXER

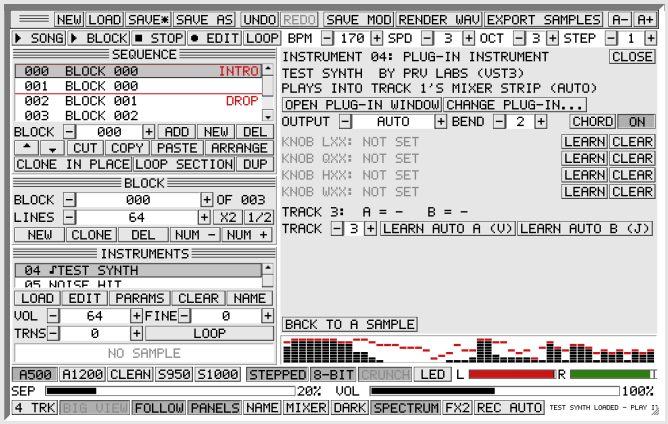
On any insert slot's effect page press `PLUG-IN...`. The first time, it looks for the plug-ins on your Mac, each one in a separate hidden copy of the app, so one that crashes can't take Paula's Revenge down. Double-click one to load it; `OPEN PLUG-IN WINDOW` shows its own controls. Its settings are saved in the song. If a song uses a plug-in this Mac doesn't have, the slot shows `!NAME`, passes the audio through and keeps the settings.

INSTRUMENTS

Pick an instrument, press `PARAMS`, then `PLUG-IN / MIDI OUT...` and `A VST3 / AU INSTRUMENT`.



PLUG-IN OR MIDI OUT?



A PLUG-IN INSTRUMENT'S PAGE

- `EDIT` opens the plug-in's own window. The laptop keys keep playing it while that window is in front.
- `OUTPUT` picks the mixer strip it plays into; AUTO is the first track that plays it. `BEND` should match the plug-in's own pitch bend range, so slides and vibrato land in tune.
- Notes play one at a time per track. `Cxx` is velocity, pitch commands become pitch bend, `Oxx` is the mod wheel, and `ECx` `EDx` `Nxx` `Kxx` `Mxx` `E9x` `Rxy` work. Sample-only commands are ignored.
- `CHORD`: `0xy` on a note's line plays a held chord, x and y semitones above the note (`037` minor, `047` major).

- LEARN next to L Q H W, then move a knob in the plug-in's window: those commands then move that knob on this instrument's notes.

AUTOMATION

Every track has two automation knobs, **AUTO A** (`Vxx`) and **AUTO B** (`Jxx`). Point one at any effect control and the pattern moves it.

1. Open an insert's effect page in the mixer. Pick the track in the `TRACK` box.
2. Press `LEARN AUTO A (V)`, then move the control: drag a built-in effect's slider, or turn a knob in the plug-in's window.
3. In the pattern, `V00` is the control's lowest and `VFF` its highest. The value holds until the next one.
4. Put `Yxx` before or beside the `V` to glide there over xx ticks instead of jumping.

- **On and off:** press LEARN, then click the effect's `BYPASS` (or an instrument's `ON`) instead. `V00` then switches it off and `V01`–`VFF` on.
- The channel page lists what each knob controls under AUTOMATION KNOBS, with CLEAR buttons. The help strip and the command list name it too.

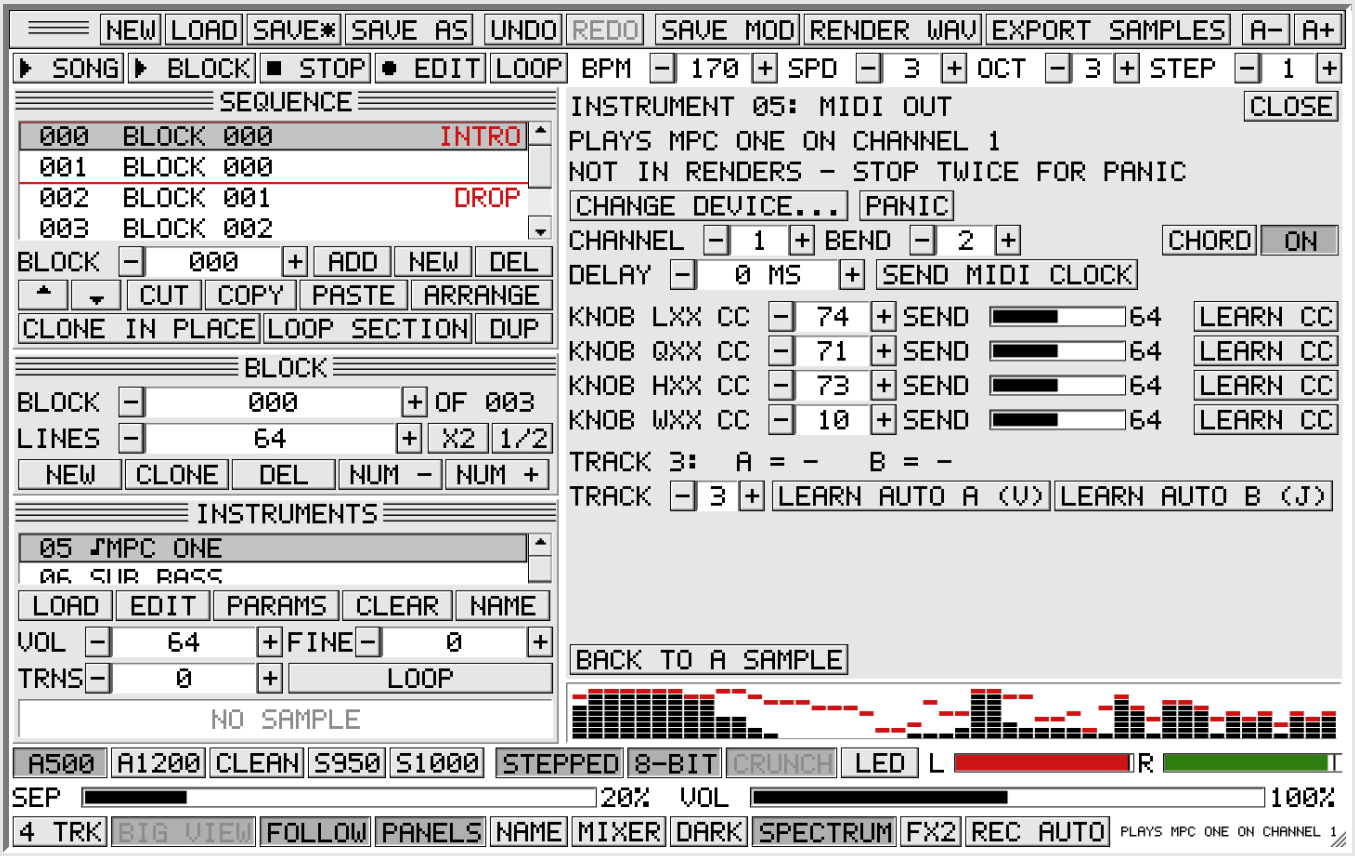
RECORDING BY HAND: REC AUTO

Switch on `REC AUTO` on the bottom row, turn EDIT on and play. Moving a learned control writes its command on the line that's playing: `V` or `J`, or `L` `Q` `H` `W` for an instrument's own knobs on the track playing it.

- It goes in the first effect column if that's free, otherwise in FX2. It never overwrites other commands; that line is skipped and the status line says so.
- One value per line per knob, and only when it changes. What the pattern plays back isn't recorded again.
- A whole pass, until you stop, is one undo step.

MIDI OUT

An instrument can play your hardware (an MPC, a microKORG, an NTS-1) over MIDI. Press PARAMS, PLUG-IN / MIDI OUT..., MIDI OUT TO HARDWARE, then double-click a MIDI output.



A MIDI-OUT INSTRUMENT'S PAGE

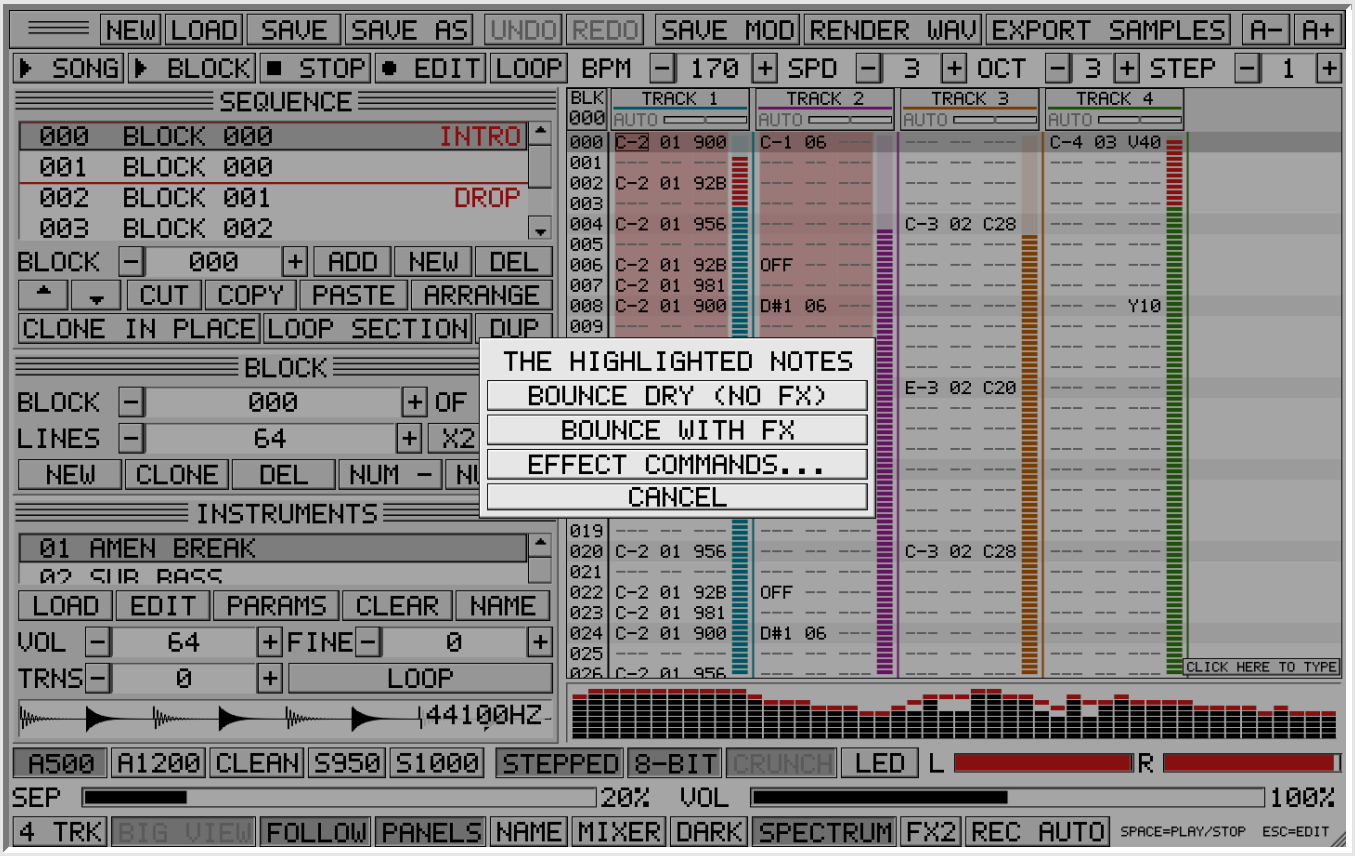
CONTROL	WHAT IT DOES
CHANNEL	MIDI channel 1-16
BEND	The hardware's pitch bend range, so pitch commands land in tune
CHORD ON	As for plug-in instruments
DELAY	-50 to +200 ms to line the hardware up with the audio. Positive: the MIDI goes later. Negative: all the audio goes later instead.
SEND MIDI CLOCK	24-a-beat clock with start, stop, continue and song position (4 lines = a beat), so the hardware follows the song

CONTROL	WHAT IT DOES
KNOB L Q H W	The CC each sends (74, 71, 73, 10 to start). LEARN CC takes the number from the next knob you turn on a controller; the SEND bar sends it now (and REC AUTO records it).
PANIC	All notes off on every MIDI output. STOP pressed while already stopped does the same.

Notes, velocity, pitch bend, the mod wheel and the timing commands work as for plug-in instruments. If the device isn't connected, the instrument shows !NAME, sends nothing and keeps its settings; CHANGE DEVICE... and the same device again reconnects it. MIDI-out instruments aren't in renders or bounces (they're hardware).

BOUNCING

Turn any part of the pattern into a new sample. Highlight some notes (they can cover several tracks), then right-click inside the highlight, or press CMD B.



RIGHT-CLICK INSIDE A HIGHLIGHT

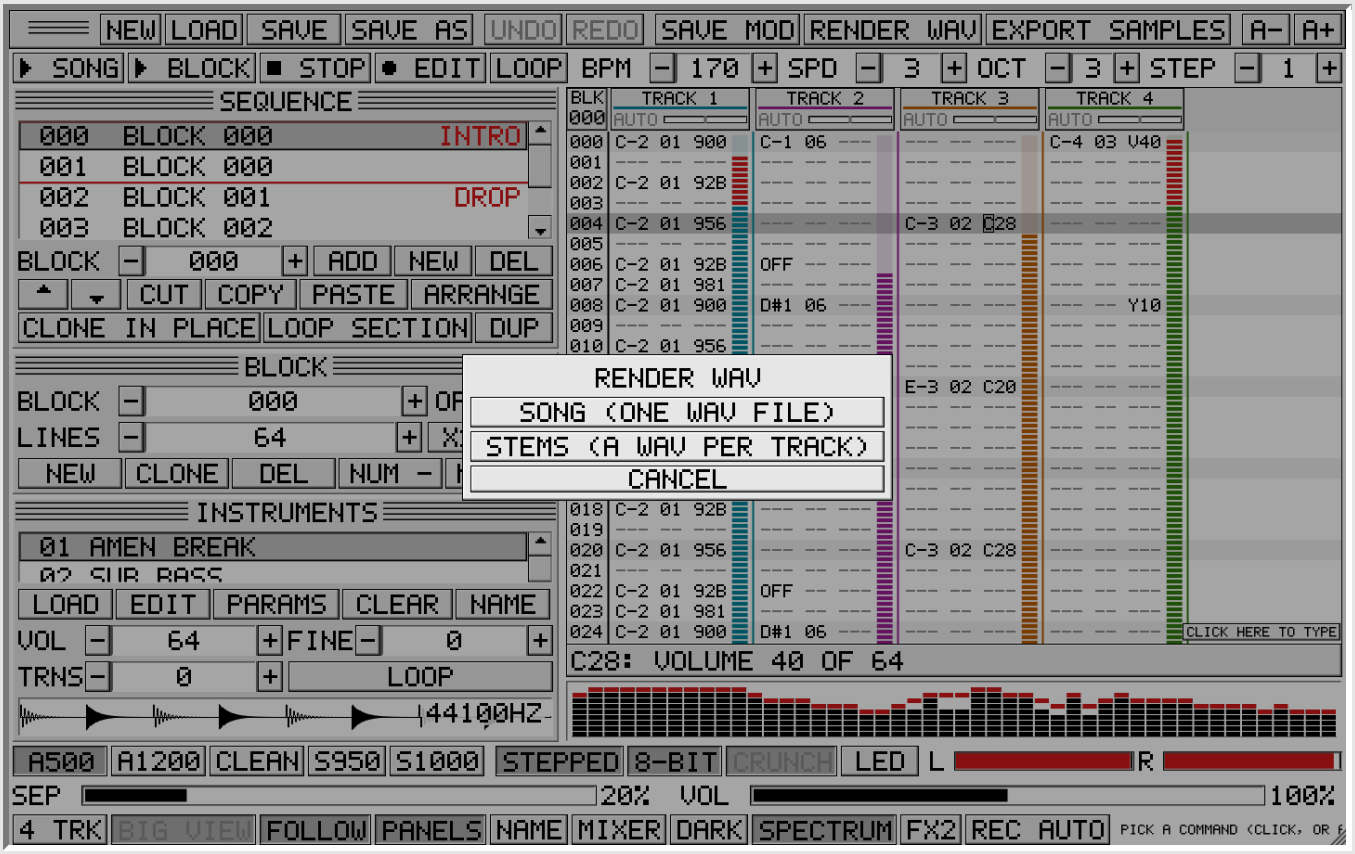
CHOICE	WHAT'S IN THE SAMPLE
BOUNCE DRY (NO FX)	The instruments' own sound, before the tracks' mixer strips. Use it when the sample will play through a strip with the same effects.
BOUNCE WITH FX	After the tracks' strips (EQ, inserts, fader) plus what they send to SEND A / B, so reverb and delay tails are in it. Not the master or the Paula / Akai output stage.

- Only the highlighted lines of the highlighted tracks play, from the first line, at the tempo and speed set there. Plug-in instruments are included.
- It records on after the last line until the sound has died away (up to 10 seconds), so tails aren't cut.

- The sample goes into the first empty instrument slot, named like `BOUNCE T1-2 000-015`, at 44.1 kHz. `C-3` plays it exactly as recorded.
- Nothing else changes: your pattern and the instruments that played it stay as they are. Undo removes the new instrument.

RENDERING

`RENDER WAV` asks what to render:



SONG OR STEMS

- **SONG:** the song once through as a 24-bit stereo WAV, mixer and all.
- **STEMS:** a folder next to the name you pick, `<name> STEMS`, with one 24-bit WAV per used track (after its strip), SEND A, SEND B and the master: `01 TRACK 1.wav` … `11 MASTER.wav`. They start together and are the same length, so they drop straight into Ableton.
- After the song ends it keeps going until everything is quiet for a second (up to 20 seconds), so tails aren't cut. Plug-in delay is taken off the start.
- MIDI-out instruments can't be rendered; the status line says which were left out.

EXPORT SAMPLES writes every instrument's sample to a folder as 01 NAME.wav, 02 NAME.wav… SAVE MOD writes a ProTracker MOD (4 tracks M.K., 8 tracks 8CHN) and says what a MOD can't hold.

SAVING & FILES

- SAVE (CMD S) writes straight to the song's file; the first time it asks for a name. SAVE AS (CMD SHIFT S) always asks, for versions. SAVE* with a star means unsaved changes.
- A .prv song holds everything: every sample (stereo too), the mixer, plug-in settings, automation, MIDI-out settings. You can move or delete the original WAVs.
- LOAD also opens ProTracker-style .mod files (4, 6 and 8 channels) and MED / OctaMED MMD0 / MMD1 modules.
- Songs saved by older versions always load.

KEYS

KEY	ACTION
Z S X D C …	Notes, lower octave (, L . ; / carry on up)
Q 2 W 3 E …	Notes, upper octave
1	Note off
SPACE	Play / stop the block · with SHIFT, the song
ESC	Edit mode on / off
ENTER	Play block · SHIFT song · CMD block from the cursor
Arrows	Move · SHIFT ↑ ↓ 4 lines · SHIFT ← → track
TAB	Next track · with SHIFT, previous
[ ]	Octave down / up
- =	Previous / next instrument
DELETE	Clear the highlight or the field · SHIFT removes the line · ALT inserts one
CMD C X V	Copy / cut / paste the highlight, or the track (SHIFT: the block)
CMD A	Highlight the whole block
CMD B	Bounce the highlight to a new instrument
CMD ↑ ↓	Transpose the track a semitone (SHIFT: an octave)
CMD Z	Undo · with SHIFT, redo
CMD [ ]	Loop start / end at the cursor
CMD S	Save · with SHIFT, save as
CMD 2 / 1	Mixer / pattern
ALT ↑ ↓	Move the selected song positions